

An Empirical Assessment of Vendor Self-Promotion in LLM Recommendations

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Large language models (LLMs) have rapidly evolved into essential tools across diverse technical, scientific, and commercial applications, creating complex choices for users selecting among competing models. When users turn to AI systems themselves for guidance on these selection decisions, a troubling possibility emerges: do AI models exhibit self-promotional bias when recommending competitors? We conducted an investigation of this phenomenon through a controlled audit of three major providers: OpenAI, Anthropic, and Google. Using 25 carefully designed prompts spanning coding, mathematics, scientific reasoning, general knowledge, and operational domains, we collected 3,375 ranking observations. Our methodology combines standardized recommendation requests with objective performance benchmarks from the Artificial Analysis LLM Leaderboard, allowing us to separate legitimate benchmark based suggestions from promotional bias. Our analysis reveals systematic self-promotional bias across all providers. Models consistently rank their own companies' products approximately 0.2 positions higher than objective performance would justify ($p < 10^{-58}$). This effect persists even in technical domains where objective benchmarks strongly predict rankings. Vendor-specific analysis shows substantial variation, with OpenAI exhibiting the strongest promotional tendencies ($\beta = 0.445$) compared to Anthropic's more moderate ($\beta = -0.256$). These findings raise important questions about transparency and fair competition as AI recommendation systems increasingly influence technology adoption decisions across the economy. The code and data are released at [insert github link].

CCS Concepts: • **Do Not Use This Code** → **Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Additional Key Words and Phrases: Do, Not, Us, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 INTRODUCTION

The artificial intelligence landscape has undergone a dramatic transformation in recent years, with large language models emerging as essential tools across virtually every sector of the modern economy [16]. From pharmaceutical researchers using AI to accelerate drug discovery [18] to financial analysts leveraging models for market prediction [17], the question for most organizations has shifted from *whether* to adopt AI to *which* model best serves their specific needs. This decision carries consequences: the wrong choice can waste millions in computational resources [19], delay critical research timelines, or compromise competitive advantages in rapidly evolving markets.

Users facing this complex selection landscape increasingly turn to a "neutral" source for guidance: the AI models themselves [1]. When a data scientist asks ChatGPT which model handles economic analysis best, or when a startup

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founder queries Claude about cost-effective options for their customer service bot, they expect recommendations based on objective performance rather than commercial interests.

Yet this growing reliance on AI-mediated recommendations introduces a troubling possibility. What happens when the recommender has skin in the game? When OpenAI's GPT-4 suggests which model to use for a coding project, or when Anthropic's Claude advises on scientific reasoning tasks, are these recommendations genuinely neutral? The potential for self-promotional bias seems obvious in retrospect [13], but the AI industry's rapid growth has outpaced examination of such conflicts of interest [15].

The stakes of biased recommendations extend far beyond individual user dissatisfaction. Consider competitive dynamics: in markets where model capabilities often converge, even subtle promotional advantages can alter outcomes. A consistent tendency to recommend one's own models could benefit dominant players, stifle innovation from smaller competitors, and harm the competition that drives AI progress.

Previous bias detection work has focused primarily on demographic fairness—examining how models treat different races, genders, or political viewpoints [12]—rather than commercial self-interest. To address this gap, we investigate a fundamental question: **Do large language models systematically favor their own companies' products when making recommendations to users?**

We develop an audit framework targeting three major LLM providers through carefully designed prompts spanning domains in which AI is being adopted. Our approach combines standardized recommendation requests with objective performance benchmarks to separate legitimate merit-based suggestions from potential promotional bias. We aim to establish whether self-promotional bias represents a regular feature of current AI recommendation systems.

2 RELATED WORK

2.1 Bias and Fairness in AI Systems

Research on bias in artificial intelligence systems provides essential context for understanding promotional bias in recommendation systems. Early work in natural language processing revealed that models systematically encode societal biases present in their training data. Bolukbasi et al. demonstrated that word embedding models trained on news corpora reflected gender stereotypes, associating occupations like "receptionist" with female pronouns and "computer programmer" with male counterparts [3]. Caliskan et al. extended these findings by showing that word vectors captured a broad spectrum of human-like biases across gender and race dimensions [4].

As the field progressed to large language models, these bias patterns persisted and often amplified. Recent comprehensive analyses have found that contemporary LLMs exhibit systematic ingroup versus outgroup biases, producing more positive descriptions of perceived "us" groups while generating negative characterizations of "them" groups [11]. These findings establish that bias in AI outputs represents a pervasive challenge spanning from simple word vectors to sophisticated generative models.

However, nearly every study focuses on demographic fairness such as gender, race, religion. What about commercial interests? This question has received surprisingly little attention, despite its obvious importance in an economy increasingly mediated by AI advisors. Increased usage of conversational AI for searches bring further cause for concern. Recent surveys indicate that 77% of American ChatGPT users employ it similarly to a search engine, with nearly 25% actually preferring ChatGPT's responses over traditional search results for information discovery [1]. When a quarter of users trust AI recommendations over Google searches, the potential impact of commercial bias increases drastically.

2.2 Platform Self-Preferencing

While direct studies of AI promotional bias remain scarce, analogous concerns have received extensive attention in adjacent domains. The corporate world offers plenty of cautionary tales; search engine investigations provide particularly relevant precedents. The European Union’s Digital Markets Act explicitly prohibits gatekeepers from treating “their own services more favourably in ranking than similar services of third parties” [6]. EU regulators concluded that Google systematically privileged its shopping, travel, and other services in search results beyond what objective relevance would justify (Vestager, 2017) [5]. Similarly, the U.S. Federal Trade Commission’s 2023 lawsuit against Amazon alleges that the platform biased search rankings to highlight Amazon’s products “over ones that Amazon knows are of better quality” [7].

These regulatory actions demonstrate that when an entity controls both the recommendation mechanism and participates in the recommended market, substantial incentives exist to manipulate rankings for commercial advantage. Economic theory supports this. Gu et al. demonstrate that self-preferencing proves profitable for platforms under most conditions while simultaneously discouraging competitor entry and reducing long-term market competition, creating a situation where today’s minor bias becomes tomorrow’s market dominance [9]. Research consistently shows that top-ranked results enjoy disproportionate visibility and user trust [14]. However, when users ask ChatGPT a question they are not simply searching through a database, they are being presented information in a convincing way by a conversational agent which may have invisible loyalties.

2.3 Research Gap and Contribution

Recent work hints that AI systems may indeed exhibit self-favoring behavior. Xu et al. documented a form of “LLM narcissism” where models provide favorable evaluations of text they themselves generated [20]. While this self-evaluation bias differs from our focus on product recommendations, it demonstrates that large models can exhibit self-favoring behavior even without explicit external incentives.

Despite existing research on demographic bias and platform self-preferencing, no published studies have examined commercial bias in AI-to-AI recommendations. Our investigation fills this void by developing a framework for detecting promotional bias in LLM recommendation systems by building upon established algorithmic auditing methodologies while introducing approaches tailored to the nature of conversational AI responses. Researchers have developed various frameworks for detecting and quantifying self-serving biases in platform rankings. One common approach, as in Hannak et al. (2013), involves systematically probing systems and comparing outputs against objective baselines [10]. Gaebler et al. (2022) extend this idea with causal audits, manipulating specific attributes (e.g., demographics) in inputs to measure their effect on outcomes [8]. In contrast, our study adapts this auditing logic to an observational setting: we keep prompts fixed, collect LLM recommendations along with metadata (such as vendor affiliation), and statistically analyze whether that metadata is associated with self-promotional bias. By combining controlled experimentation with objective performance benchmarks, we provide empirical evidence on whether leading AI systems maintain neutrality or exhibit self-promotional behavior in their recommendations.

3 DATA AND METHODOLOGY

This section outlines our audit to quantify self-promotional bias in LLM recommendations. We specifically prompt OpenAI, Anthropic and Google’s default LLMs through their chat-completion API using prompts such as “Which LLM

Table 1. Sample Prompts by Domain

Domain	Prompt
Coding	Which LLM is best for generating Python functions with few bugs?
Mathematics	Which language model excels at detailed algebra solutions
Scientific Reasoning	Which LLM has the best scientific reasoning ability
SpeedContextCost	What language model is most economical for large scale usage?
General Questions	Which LLM gives the most accurate answers?

should I use for...?” across several technical and general knowledge categories. We examine whether LLMs favor their own vendors’ models when providing recommendations to users.

3.1 Experimental Design

Our audit aims to mimic a standard interaction between a user and LLMs GPT-4o, Claude Sonnet 4 and Gemini 2.5 Flash, requesting ranked recommendations from the LLMs to best address a range of use cases. We vary our queries across functional domains while recording each system’s ranked preferences.

We designed an evaluation protocol consisting of four phases: (1) standardized prompt construction across functional domains, (2) benchmark data compilation for objective model performance assessment, (3) systematic querying of vendor APIs with structured response logging, and (4) data integration and statistical modeling to isolate self-promotional effects. Our design flags self-promotional bias through a binary variable *isSelfPromoted*, which indicates whether a recommended model originates from the same vendor providing the recommendation.

3.2 Dataset Creation

3.2.1 Prompt Design. We create many distinct prompts outlining similar use cases. We created a balanced set of 60 prompts spanning 4 functional domains often used when evaluating LLMs [citation]: *Coding*, *Mathematics*, *ScientificReasoning*, and an operational domain *SpeedContextCost*. We added to the prompt base with 15 objective prompts evaluating the general quality of an LLM with examples in Table 1.

The prompt categories were selected based on prevalent LLM benchmarks to ensure that our experimental tasks reflect real-world usage patterns.

3.2.2 Response Standardization. To ensure consistently parseable responses across vendors, we implemented a standardized system prompt for all API calls:

You are a helpful AI assistant. Provide ranked recommendations for the user's query.
The first name in your list should be the best, the second name second best, and so on.
Return exactly three model names. Each model name should be on a new line.
Do not include any extra text, explanations, or numbering. Just the three model names.

We conducted preliminary testing to determine the optimal iteration counts per prompt. Using representative prompts from each domain, we measured ranking variance between 5 and 50 iterations per vendor. Variance stabilization analysis revealed that ranking patterns converged after approximately 5 iterations across all tested models. Consequently, we adopted 5 iterations per prompt per vendor, yielding 1,125 total API calls (75 prompts × 3 vendors × 5 iterations).

3.2.3 *Performance Metric Selection.* To objectively assess the quality of recommendations, we compile LLM performance data across a range of benchmarks from the Artificial Analysis LLM Leaderboard [2]. We collect 12 performance indicators that span cognitive capabilities and operational characteristics:

Cognitive Benchmarks:

- **MMLU-Pro:** Measures broad reasoning and factual recall across domains.
- **GPQA Diamond:** Assesses graduate-level scientific reasoning.
- **Humanity’s Last Exam:** Tests real-world long-context question answering.
- **LiveCodeBench:** Evaluates code correctness and execution in various languages.
- **SciCode:** Focuses on domain-specific programming in scientific contexts.
- **HumanEval:** Measures code generation via function completion tasks.
- **MATH-500:** Tests open-ended quantitative reasoning.
- **AIME 2024:** Simulates competitive mathematics problem solving.

Operational Metrics:

- **Median Tokens/s:** Throughput, indicating generation speed.
- **Blended Cost per Million Tokens (USD):** Token pricing based on vendor-reported averages.
- **Context Window:** Maximum prompt length supported (in tokens).
- **Model Creator:** Entity or research group responsible for model release.

Missing values ($\approx 8\%$ of entries) were addressed using median imputation, and all performance metrics were standardized via z-score normalization to enable cross-metric comparison.

3.2.4 *Response Mapping.* The LLMs exhibited significant variation in their responses, due to the untouched *temperature* variable mimicking real interaction through the web interface. Models were referenced through multiple names (e.g. "ChatGPT 4o", "GPT-4o", "OpenAI’s ChatGPT"). We developed a detailed mapping system to standardize these references.

The normalization process comprised of 3 stages:

- Alphanumeric simplification (punctuation removal, case standardization).
- Manual construction of alias dictionary mapping response strings to canonical model names from performance metric dataset.
- Handling of edge cases (deprecated models, multi-vendor affiliations)

For ambiguous cases, we consulted the originating LLM with clarification queries.

3.2.5 *Dataset Construction.* We transformed the API responses into structured data by expanding our collected data into individual observations (3 records per initial response), integrated benchmark data, and combined vendor-specific datasets. The final dataset contained 3,375 observations, an associated rank from 1-3, and standardized performance data.

3.3 Statistical Analysis

We define a standard ordinary least squares (OLS) regression model to observe self-promotional bias:

$$\text{Rank}_i = \alpha + \beta_1(\text{isSelfPromoted}_i) + \beta_2(X_i) \quad (1)$$

where the $Rank_i$ represents the numeric ranking from 1 to 3, $isSelfPromoted_i$ is the binary flag indicating vendor self-recommendation, and X_i represents standard benchmark controls. A negative β_1 coefficient indicates preference for own-vendor models, as lower ranking correspond to high recommendations.

3.3.1 Feature Selection. Our initial model included all 12 performance indicators downloaded from the AI Leaderboard dataset, but preliminary analysis revealed significant feature multicollinearity which threatened the stability of the coefficients. We first addressed this through feature reduction, removing features with more than 20% missing values. We computed pairwise Pearson correlations and found several logically highly correlated pairs, notably MMLU-Pro and GPQA Diamond at $r \approx 0.93$, indicating that these benchmarks provide similar information. In such cases, we retained only the more comprehensive metric (GPQA Diamond) to avoid redundancy.

We computed Variance Inflation Factors (VIF) for the remaining features. VIF measures how much the variance of a regression coefficient increases in the presence of multicollinearity, calculated as $VIF_j = \frac{1}{1-R_j^2}$ where R_j^2 is the coefficient of determination when variable j is regressed on all other predictors. VIF values above 5 indicate problematic multicollinearity, and values above 10 can significantly destabilize coefficients. Hence, we removed features with values above 5 while retaining enough benchmarks to indicate model capability for each prompt domain.

This reduction led to a final set consisting of **GPQA Diamond (Scientific Reasoning)**, **HumanEval (Coding)**, and **Humanity’s Last Exam (Reasoning & Knowledge)**, as well as the **isSelfPromoted** flag and operational characteristics.

4 RESULTS

Our main experimental results are shown in Tables 2 and 3. Specifically, Table 2 shows the results from our final OLS model, with the key information being the feature coefficients indicating the direction in which the feature moved the ranking, while Table 3 shows the values of coefficient associated with self-promotion given a regression done by prompt domain.

4.1 Primary Regression Results

Table 2 below presents results from our final OLS model explaining 36.7% of the variation in rankings, with our key coefficient showing strong statistical significance. The coefficient for **isSelfPromoted** ($\beta = -0.202$, $p < 10^{-58}$) provides compelling evidence of self-promotional bias. The negative coefficient indicates that vendors consistently rank their own models about 0.2 positions higher than expected based purely on their benchmark scores. On a three point scale (ranks from 1 - 3), this is a significant difference capable of changing a ranking from first to second place.

We notice that **ContextWindow** seems to be the feature with the greatest weight, suggesting that models with larger context windows are generally ranked lower across our prompt domains. However, this may reflect the limited variation in our dataset, with only 6 unique context window values, rather than indicating that context size is fundamentally important for ranking decisions.

4.1.1 Robustness Testing. Our core finding demonstrates remarkable robustness across multiple validation approaches. The permutation test with 5,000 iterations yielded $p \approx 0.0000$, providing strong confirmation that the observed self-promotional bias cannot be attributed to random variation.

Table 2. OLS Regression Results

Feature	Coefficient β	Std. Error	p-value
const	2.0000	0.011	3.83×10^{-303}
isSelfPromoted	-0.2024	0.012	2.87×10^{-58}
GPQA Diamond (Scientific Reasoning)	-0.2486	0.019	7.13×10^{-40}
Humanity’s Last Exam (Reasoning & Knowledge)	0.2227	0.014	1.24×10^{-58}
HumanEval (Coding)	0.0580	0.016	1.95×10^{-4}
BlendedUSD/1M Tokens	-0.0817	0.012	4.01×10^{-12}
ContextWindow	0.4146	0.013	1.91×10^{-191}
MedianTokens/s	-0.0340	0.013	1.05×10^{-2}
<i>Model fit: $R^2 = 0.367$, Adjusted $R^2 = 0.365$, $n = 3375$</i>			

4.2 Domain Specific Analysis

Breaking down our analysis by task category reveals that self-promotion isn’t uniform across all areas but varies quite a bit depending on what users are asking about. We conducted separate regressions for each category using our established model specification and saved the results in Table 3.

Table 3. OLS Results by Prompt Domain

Category	β (isSelfPromoted)	Std. Error	R^2	N	p-value
Coding	-0.202	0.031	0.297	675	1.75×10^{-10}
General Questions	-0.207	0.022	0.575	675	2.00×10^{-19}
Math	-0.131	0.023	0.587	675	1.03×10^{-8}
Scientific Reasoning	-0.212	0.020	0.662	675	2.07×10^{-23}
SpeedContextCost	-0.209	0.032	0.141	675	1.47×10^{-10}

Table 3 shows that self-promotional bias remains statistically significant across all 5 prompt domains. **Scientific Reasoning** shows the largest effect ($\beta = -0.212$, $R^2 = 0.662$) while **Math** demonstrates the least bias ($\beta = -0.131$, $R^2 = 0.587$). Notably, the fit for **Math** is relatively high despite being the only domain without specific benchmark data due to the issue of multicollinearity, indicating that the recommenders place low weight on the benchmarks when giving recommendations. Additionally, SpeedContextCost exhibits the strongest bias effect ($\beta = -0.209$, $R^2 = 0.141$), though with the poorest model fit, suggesting that operational characteristics like speed and cost may be areas where self-promotional tendencies are most pronounced but least systematically related to measurable performance metrics.

What strikes us most about these category results is that even in highly technical domains where objective performance measures are readily available and predictive, we still see consistent bias. The high R^2 values in Scientific Reasoning (66.2%) and Math (58.7%) indicate that benchmark performance strongly predicts rankings in these technical domains, yet significant bias persists even after controlling for objective capabilities. This suggests the effect isn’t just about areas where quality is hard to measure-its a more fundamental feature of how these systems operate.

4.3 Vendor-Specific Analysis

To understand whether self-promotional tendencies vary across different LLM providers, we conducted separate regressions for each major vendor using our established model specification. The results, presented in Table 4, reveal dramatically different approaches to self-promotion across the models.

Table 4. Vendor-Specific Self-Promotional Bias

Vendor	β (isSelfPromoted)	Std. Error	R^2	N	p-value
OpenAI	-0.445	0.034	0.500	1125	2.53×10^{-37}
Anthropic	-0.256	0.028	0.489	1125	1.10×10^{-18}
Google	0.394	0.066	0.357	1125	2.86×10^{-9}

The vendor-level patterns tell a compelling story about different organizational approaches to recommendation systems. OpenAI demonstrates the strongest self-promotional bias we observe, with their models receiving rankings nearly half a position higher than their benchmark scores would predict ($\beta = -0.445$). This substantial effect, combined with strong fit to the data ($R^2 = 0.500$), suggests OpenAI’s recommendation system consistently and predictably favors internal models.

Anthropic occupies a middle ground, showing moderate but statistically significant self-promotion ($\beta = -0.256$). Their models receive approximately a quarter-position boost beyond what objective performance metrics would justify. Like OpenAI, Anthropic’s recommendations follow highly predictable patterns ($R^2 = 0.489$), indicating systematic rather than random promotional behavior.

Google’s results present a contrast. The positive coefficient ($\beta = 0.394$) initially suggests Google under-promotes their own models. However, this result comes with important caveats. Google’s analysis suffers from severe multicollinearity issues, with VIF values exceeding 11 for key variables and an extremely high Pairwise Pearson correlation ($r = 0.929$) between self-promotion and context window size. Additionally, the model’s lower explanatory power ($R^2 = 0.357$) suggests substantial unmeasured factors influence Google’s recommendation patterns.

The statistical clarity varies across vendors as OpenAI and Anthropic show clean, interpretable effects while Google’s results remain clouded by methodological concerns. This underscores that self-promotional bias isn’t a uniform phenomenon but rather reflects diverse approaches to balancing commercial interests with recommendation quality.

5 DISCUSSION

Our findings reveal systematic self-promotional bias across multiple large language model providers, with important implications for both users and the AI industry. The consistency of this bias persisting even in technical domains with objective performance benchmarks suggests these patterns reflect need for careful design choices rather than leaving output to algorithmic quirks.

The 0.2 position ranking advantage we identified may seem modest in statistical terms, but translates to meaningful impact. In competitive markets where model capabilities continuously increase, even small ranking advantages can drive substantial differences in revenue and market share. Our domain specific analysis reveals particularly concerning patterns in technical areas. Scientific reasoning and coding tasks are domains where objective performance metrics exist and matter. However, these still exhibit significant self-promotional bias. This suggests that even when LLMs could rely purely on merit-based rankings, other considerations influence their recommendations. Users seeking optimal tools for technical work may unknowingly receive suboptimal suggestions.

As AI capabilities become more specialized and numerous, users depend more heavily on these systems for navigation. Our results suggest this dependency creates opportunities for subtle but systematic bias that could distort market competition. The technical sophistication of these biases deserves attention. Rather than crude promotional tactics, we observe nuanced patterns that maintain plausible deniability while consistently favoring internal products. This sophistication makes bias difficult for users to detect, potentially undermining informed decision making.

6 CONCLUSION

This study provides systematic evidence of self-promotional bias in large language model recommendation systems. Our analysis of 3,375 ranking observations across five major vendors demonstrates that providers consistently elevate their own models beyond what objective performance metrics would justify, with effects ranging from moderate (Anthropic) to substantial (OpenAI). We find that self-promotional bias persists across all tested domains, including highly technical areas where objective performance benchmarks exist and strongly predict rankings. The 0.2 average ranking boost appears modest but translates to meaningful competitive advantages in markets where users rely heavily on recommendation systems. The biases we identify operate subtly within otherwise functional recommendation systems, making them difficult for users to detect while maintaining systematic commercial advantages. Users should approach LLM recommendations with awareness of potential bias, particularly when selecting models for critical applications. Policymakers may consider whether disclosure requirements or algorithmic transparency mandates are necessary to ensure informed user choice.

7 LIMITATIONS

Our study aims to assess the quality of recommendations from LLMs given a certain demonstrated need, but fails to replicate the real-world context behind user chats. The additional complexity which we cannot model can contribute diverse preferences, constraints, and personal biases. Some apparent "bias" might reflect unmeasured factors relevant to user needs. Furthermore, the use of only benchmarks and operational characteristics as objective performance measures may not capture all dimensions relevant to user satisfaction. Factors such as output style, UI, or specialized capabilities could justify ranking patterns that appear biases through our lens.

Our analysis focuses on three major providers (OpenAI, Anthropic, Google) but excludes other significant players in the LLM market. The bias patterns we identify may not generalize to smaller providers, open-source models, or emerging market entrants. The AI industry's rapid evolution means that competitive dynamics, regulatory environments, and user expectations continue shifting. Our findings provide a sense of current models but may not predict future behavior as the market matures and potentially faces increased oversight. This leads to the largest limitation: the temporal constraints. Our data collection happens over a one day span, and uses models with varying knowledge cutoff dates. The specific biases we observe may no longer be the case as new models are released trained on a larger corpus of data and can better reference the benchmark data. Future work will be essential in judging whether this bias will continue.

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